Gaussian HMM Regime Model — Cheat Sheet

MARTA — Layer 1: Regime Discovery (`regime/hmm.py`)

1. Model definition

Hidden regime $z_t \in \{1, \dots, K\}$, observed feature vector $x_t \in \mathbb{R}^d$ ($d = 14$: ATR, ADX, RSI, BB width, MACD hist., OBV, vol z-score, realized vol (z-scored & raw), vol ratio, return skew/kurtosis, autocorr(1), close return).

- **Initial dist.:** $\pi_k = P(z_1 = k)$
- **Transition matrix:** $A_{jk} = P(z_{t+1} = k \mid z_t = j)$
- **Emission (Gaussian, diag. cov.):**

$$P(x_t \mid z_t = k) = \mathcal{N}(x_t; \mu_k, \Sigma_k)$$
$$= \prod_{i=1}^{d} \frac{1}{\sqrt{2\pi\sigma_{k,i}^2}} \exp\left(-\frac{(x_{t,i} - \mu_{k,i})^2}{2\sigma_{k,i}^2} \right)$$

Diagonal Σ_k: features independent within a state — avoids one state's full covariance "running away" unconstrained during EM.

2. E-step — Forward–Backward

Forward (probability of the past & current state):

$$\alpha_1(k) = \pi_k \mathcal{N}(x_1; \mu_k, \Sigma_k)$$
$$\alpha_t(k) = \left[\sum_j \alpha_{t-1}(j) A_{jk} \right] \mathcal{N}(x_t; \mu_k, \Sigma_k)$$

Backward (probability of the future given current state):

$$\beta_T(k) = 1$$
$$\beta_t(k) = \sum_j A_{kj} \mathcal{N}(x_{t+1}; \mu_j, \Sigma_j) \beta_{t+1}(j)$$

State posterior (soft belief — *this is the* `regime_prob` *vector fed downstream*):

$$\gamma_t(k) = P(z_t = k \mid x_{1:T}) = \frac{\alpha_t(k)\, \beta_t(k)}{\sum_j \alpha_t(j) \beta_t(j)}$$

Pairwise transition posterior:

$$\xi_t(j,k) = \frac{\alpha_t(j) A_{jk}\, \mathcal{N}(x_{t+1}; \mu_k, \Sigma_k)\, \beta_{t+1}(k)}{P(x_{1:T})}$$

3. M-step — parameter re-estimation

$$\pi_k = \gamma_1(k), \qquad A_{jk} = \frac{\sum_t \xi_t(j,k)}{\sum_t \gamma_t(j)}$$
$$\mu_k = \frac{\sum_t \gamma_t(k)\, x_t}{\sum_t \gamma_t(k)}, \qquad \sigma_{k,i}^2 = \frac{\sum_t \gamma_t(k)(x_{t,i} - \mu_{k,i})^2}{\sum_t \gamma_t(k)}$$

E-step then M-step = one **Baum–Welch / EM** iteration. Log-likelihood is non-decreasing every iteration (EM guarantee); stop at `tol=1e-4` or `max_iter=200`. Best-of-**5 seeds** kept (EM finds only a local optimum — multi-restart guards against a bad init).

4. Model selection — BIC

$$\mathrm{BIC}(K) = -2 \ln \hat{L}_K \; + \; p_K \ln n, \qquad p_K \approx K^2 + 2Kd$$

- K^2: transition matrix free params; $2Kd$: per-state mean (d) + diagonal variance (d), for K states.
- Grid search $K \in \{2, \dots, 8\}$, diagonal cov, `tol=1e-3`, 100 iters.
- Pick $\arg\min_K \mathrm{BIC}(K)$; **flat-curve guard** — if a larger K improves BIC by $< 5\%$ of the best value, keep the smaller K.
- Result on the crypto5 panel: $K = 6$.

5. Decoding (after fitting)

- **Viterbi:** $\arg\max_{z_{1:T}} P(z_{1:T} \mid x_{1:T})$ — single most likely *whole path* → `regime_label`.
- **Marginal posterior** $\gamma_t(k)$ at each t → `regime_prob_0..K-1` — the vector actually consumed by the agents / RL observation. *Prefer this over the hard label.*

6. Project-specific numbers to cite

- $d = 14$ features; 720h rolling z-score, `min_periods=168` (except `realized_vol_raw`, `close_return` — kept unnormalized to preserve absolute crash magnitude).
- Sequence boundaries: `lengths=` passed to hmmlearn so multi-asset concatenated training data isn't treated as one continuous chain.
- Semantic validation: pass iff $\mathrm{std}_k(\mu_{k,\text{realized_vol}}) > 0.001$ — regimes must differ in average volatility, not just be arbitrary clusters.

Honest limitation (state proactively): regime label vs. forward price direction agreement $\approx 49\%$ (coin flip, zigzag-leg test, 111 legs). Regimes cluster on *volatility/trend-magnitude*, not directional bias — expected, since the HMM clusters the joint feature distribution, not forward return sign.